

Jerin Romijah Tuli

AI/ML Researcher - NLP, LLMs & Biomedical Signal Processing

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OBJECTIVE

Seeking a fully funded MS/PhD position in artificial intelligence and machine learning research, with a focus on natural language processing, LLM evaluation, and deep learning for biomedical signals. My work centers on making model outputs trustworthy under uncertainty whether that means an LLM recognizing an unanswerable question in a retrieval-augmented pipeline, or a classifier correctly hedging on an ambiguous EEG epoch at a sleep-stage boundary.

EDUCATION

2022 – 2026	B.Sc. in Computer Science and Engineering Rajshahi University of Engineering & Technology (RUET), Bangladesh — CGPA 3.78/4.00 <i>Thesis: Automated EEG Sleep Stage Classification. Supervisor: Md. Nasif Osman Khansur.</i>
2020 – 2021	Higher Secondary Certificate (HSC), Science Viqarunnisa Noon School & College - GPA 5.00
2018 – 2019	Secondary School Certificate (SSC), Science Viqarunnisa Noon School & College - GPA 5.00

PUBLICATIONS (PEER-REVIEWED)

1. J. R. Tuli, M. S. A. Pritom, T. N. H. Naem. “*MindFlayer at SemEval-2026 Task 8: DUALRAG - Answerability-Aware Generation for Multi-Turn RAG Conversations.*” Proc. 20th International Workshop on Semantic Evaluation (SemEval 2026), ACL Anthology, pp. 2314–2321.
Zero-shot answerability-routing system built on Llama-4-Scout-17B (Groq API); ranked 8th of 26 teams with 100% refusal accuracy on unanswerable questions, outperforming a GPT-OSS-120B baseline.
2. J. R. Tuli, T. N. H. Naem, M. S. A. Pritom. “*MindFlayer at SemEval-2026 Task 13: LACR-ENS - Calibration-Aware Ensemble Routing for Cross-Language AI-Generated Code Detection.*” Proc. 20th International Workshop on Semantic Evaluation (SemEval 2026), ACL Anthology, pp. 2322–2329.
Calibration-aware ensemble (UniXCoder + GraphCodeBERT) for AI-generated code detection across 8 languages; cut out-of-distribution calibration error from 0.18 to 0.11 while improving macro-F1.

RESEARCH IN PROGRESS

Calibrated Answerability Gating for Multi-Turn RAG (CAG-MT). A confidence-gated retrieval-augmented generation system reaching 52.7% unanswerable recall at 88.6% answerable accuracy on MTRAG-842/MTRAG-UN-507, breaking a published 0.240 unanswerable-F1 ceiling. With M. S. A. Pritom.

The Social Mirage. A representation-level probing study of whether instruction-tuned LLMs retain private beliefs under social pressure; finds a persistent Representation–Mind Gap (+0.55) and a 7.7× rise in strategic conformity from base to instruction-tuned models. With M. S. A. Pritom.

Automated EEG Sleep Stage Classification. Undergraduate thesis: a deterministic spectral-enhancement module feeding per-channel vision transformers with gated cross-channel fusion; 78.77% Macro-F1 / 78.91% accuracy on a verified subject-independent split of Sleep-EDF Expanded. Being prepared for IEEE JBHI.

Bengali RAG Hallucination Evaluation Pipeline. Independent research isolating retrieval failure from generation failure in Bengali RAG; introduces a Language Gap Score against an English baseline. Targeting the MeLLM workshop, ACL 2026.

BanglishBench. 1,130-sample Bengali–English code-switching benchmark across safety classification, NLI, and QA, with a composite BBS metric. Accepted to SemEval 2026.

Covert Coordination in Cyber-Physical Multi-Agent LLM Systems. Asks whether LLM agents deliberately kept unable to communicate directly can still influence one another through a shared environment alone, using a simulated multi-agent clinical deployment as the test case, and develops a cause-and-effect detector to catch that influence without over-flagging ordinary behavior.

Dependable Dual-Utility Relief for AI Data Centers. Explores how much electricity and local water relief an AI data center can dependably commit to at the same time during heat extremes, using workload flexibility, cooling-mode switching, and storage, coordinated through an uncertainty-aware learned dispatcher.

WORK EXPERIENCE

Mar 2025	Software Development Intern <i>Brain Station 23 Limited</i> Collaborated in an agile squad on AI-driven healthcare solutions, participating in daily scrums and code reviews; contributed to AI/ML model integration and production-grade software development under senior engineering mentors.
2025	System Analyst, Group Project <i>Gizan Tech</i> Analyzed organizational workflows and data-handling processes; designed DFDs and process models for centralized data management; translated findings into technical recommendations presented to stakeholders.

SKILLS & INTERESTS

Research Interests	Natural Language Processing, LLM Evaluation, Retrieval-Augmented Generation, Deep Learning, Biomedical Signal Processing
Languages	English - Full professional proficiency; Bangla - Native
Programming	Python, SQL, C, C++, Java, LaTeX
Platforms / Frameworks	PyTorch, TensorFlow/Keras, Groq API, Flask, Django, scikit-learn

CERTIFICATIONS

DataCamp - Data Analyst Associate
DataCamp - Data Manipulation with pandas
DataCamp - Intermediate Python
Machine Learning with Python - Completion Certificate
PHP (Laravel) Certificate - EDGE-DSTS-116-6631-00001
Brain Station 23 - Internship Certificate